

Serum Fibrosis Markers in Children.

In the last chapter we measured how stiff the liver is with a probe on the skin. This chapter asks a simpler question. Can we estimate how much scar there is in a child's liver from an ordinary blood test?

The idea is attractive. Blood tests are cheap, they are available in every hospital, they need no special machine and no trained operator, and they can be repeated at every clinic visit. In adult hepatology, two blood scores in particular, called APRI and FIB four, are used every day. In many adult clinics a low FIB four is enough to reassure a patient and send them back to their family doctor.

The problem is that almost all of these scores were invented in adults with viral hepatitis. When we use them in a child, we carry over assumptions that are not true in children. Children have different normal values for liver enzymes and platelets. They grow, and growth itself turns over collagen. And they have different diseases, many of them biliary, where the blood tests behave in a different way.

The main idea of this chapter is simple. A serum score is a calculation built from a few numbers, and every one of those numbers has its own biology outside the liver. Before you trust a score in a child, ask what each number in it really measures. If you do that, you will avoid most of the mistakes that are made with serum fibrosis markers in children.

There is a second idea that runs through the whole chapter. In children, these scores are usually better at predicting what will happen to the child than at telling you the exact fibrosis stage today. The reason is the platelet count. A falling platelet count tells you that the spleen is getting bigger, and that usually means portal pressure is rising. That is useful information, even when the score cannot stage the scar.

So this chapter asks seven questions. First, what does each marker measure? Second, why does FIB four fail in children? Third, what moves APRI in a child? Fourth, what is the right blood test in biliary atresia? Fifth, where do APRI and the GGT-based score earn their place, in cystic fibrosis? Sixth, what about fatty liver and the direct matrix test called ELF? And seventh, what do the guidelines say?

A short note on the sources. The most important document is the guideline of the American liver society, AASLD, on blood-based non-invasive liver disease assessment, written by Sterling and colleagues and printed in 2025. It is the only

major guideline that addresses serum markers in all paediatric chronic liver disease. For fatty liver in children we use the AASLD paediatric guidance of 2025, written by Xanthakos and colleagues, and the ESPGHAN position paper of 2026, written by Mann and colleagues. For cystic fibrosis we use the 2024 guidance of the Cystic Fibrosis Foundation, written by Sellers and colleagues. In the United Kingdom, the NICE guideline on fatty liver, called NG forty-nine, was updated in July 2026. And for hepatitis B we use the World Health Organization guideline of 2024. As in the previous chapters, most of the paediatric numbers come from single studies, and we will name them as we go.

A note on words. Sensitivity is the proportion of children with the disease whom the test picks up. Specificity is the proportion of children without the disease whom the test correctly calls normal. The negative predictive value is the chance that a child with a negative test really does not have the disease. The area under the curve, or AUROC, is a single number between one half and one that describes how well a test separates two groups. One half is no better than tossing a coin. One is perfect. Anything above about zero point eight is considered good, and anything below about zero point seven is poor.

Fibrosis stages are written with the letter F. F zero is no fibrosis, F one and F two are mild and moderate fibrosis, F three is advanced, or bridging, fibrosis, and F four is cirrhosis. When we say F three to F four, we mean advanced fibrosis or cirrhosis.

Part one. What each marker measures.

There are two kinds of serum fibrosis markers. The first kind are indirect markers. They are calculations made from routine blood tests, such as the liver enzymes and the platelet count. They do not measure scar at all. They measure things that tend to go along with scar, such as liver cell injury and a big spleen. The second kind are direct markers. These are proteins that are made or broken down when collagen is laid down in the liver. They measure the turnover of the scar tissue itself.

Let us go through the main markers one by one.

APRI stands for the AST to platelet ratio index. You take the AST and divide it by the upper limit of normal for AST in your laboratory. Then you divide the result by the platelet count, and multiply by one hundred. So APRI goes up when the AST goes up, and it goes up when the platelets go down. It was built in 2003 by Wai and colleagues, in adults with hepatitis C. What it really sees is a combination of liver cell injury, through the AST, and portal hypertension, through the platelets.

FIB four is the second common score. You multiply the age in years by the AST. Then you divide by the platelet count multiplied by the square root of the ALT. It was built in 2006 by Sterling and colleagues, in adults with both HIV and hepatitis C. It sees the same two things as APRI, liver cell injury and portal hypertension, but then it multiplies everything by the patient's age. As we will see, that age term is the reason it fails in children.

GPR stands for the GGT to platelet ratio. It works exactly like APRI, but with GGT in place of AST. You divide the GGT by its upper limit of normal, divide by the platelets and multiply by one hundred. It was built in 2016 by Lemoine and colleagues, in adults with hepatitis B in West Africa. Because GGT comes mainly from the bile ducts, GPR sees biliary injury plus portal hypertension. In children the catch is that GGT is physiologically very high in the first months of life.

The paediatric fatty liver scores are the only ones that were built in children. The Paediatric NAFLD Fibrosis Score, or PNFS, uses the ALT, the alkaline phosphatase, the GGT and the platelets. The Paediatric NAFLD Fibrosis Index, or PNFI, uses the age, the waist circumference and the triglycerides. They were developed between 2009 and 2014. Their catch is that alkaline phosphatase rises with bone growth, and age and waist change as the child grows.

The ELF test, which stands for Enhanced Liver Fibrosis, is a direct marker. It combines three proteins: hyaluronic acid, the amino-terminal propeptide of type three procollagen, and a protein called TIMP one, which is a tissue inhibitor of the enzymes that break down collagen. We will call the propeptide of type three procollagen by its short name, P three NP. ELF was built in 2004 in adults with a mix of chronic liver diseases. It measures matrix turnover directly. The catch in children is that growth also turns over collagen, in bone and in every other tissue, so a growing child has a high ELF without any liver disease.

Finally, MMP seven, or matrix metalloproteinase seven, is a single protein measured in serum. It comes from injured bile duct cells. It was developed in the last ten years for infants with cholestasis, as a test for biliary atresia. Its catch is that the assay and the cut-off are not yet standardised between laboratories.

It helps to see how firmly the common scores are rooted in adult viral hepatitis. APRI was built on two hundred and seventy adults with hepatitis C, and in that group it had an AUROC of zero point eight nine for cirrhosis. FIB four was built on eight hundred and thirty-two adults with HIV and hepatitis C. In that group, a FIB four below one point four five had a negative predictive value of ninety percent, and a FIB four above three point two five had a specificity of ninety-seven percent. GPR was built on one hundred and thirty-five adults with hepatitis B in the Gambia, and when it was tested in France it was no better than APRI.

Even in adults, these scores have limits. McPherson and colleagues showed in 2017 that FIB four does poorly in young adults. In adults with fatty liver aged thirty-five or younger, its AUROC for advanced fibrosis was only zero point six zero. At the other end, adults over sixty-five need a higher cut-off, of two point zero. If a score bends this much across adult life, we cannot expect it to survive childhood.

Part two. FIB four and the problem of age.

The easiest way to understand why FIB four fails in children is to work through an example.

Imagine a patient with an AST of eighty units per litre, an ALT of sixty-four units per litre, and a platelet count of one hundred and twenty. These are worrying numbers. The AST is raised, and the platelets are low, which suggests a big spleen and possible portal hypertension. Now we keep these blood results exactly the same and change only the age.

At seventy years old, the FIB four is five point eight three. At forty-five years old, it is three point seven five. Both are above the adult rule-in cut-off of three point two five, so the score says advanced fibrosis is likely. At fifteen years old, the FIB four is one point two five. At eight years old, it is zero point six seven. At two years old, it is zero point one seven. And in a baby of three months, it is zero point zero two.

The fifteen-year-old is below the adult rule-out cut-off of one point three zero, which is the cut-off used for fatty liver in adults. So the same blood results that point to advanced fibrosis in a forty-five-year-old would reassure us in a teenager. The liver chemistry is the same, but the verdicts are opposite. In a child, FIB four mainly measures the birthday.

The paediatric studies confirm this. In a European study by Kalveram and colleagues, published in 2024, four hundred and fifty-seven children with fatty liver from ten centres had a liver biopsy. The AUROC of FIB four for advanced fibrosis was only zero point six one one. That is poor.

Can we fix FIB four by using much lower cut-offs in children? A study by Li and colleagues, published in 2025, tried this in infants with biliary atresia. The best cut-offs were zero point zero seven three for F two or more, and zero point one zero five for F three or more. These are about one tenth of the adult values. They have not been tested in other centres, and they would not apply to older children.

In cystic fibrosis, a study by Leung and colleagues from Houston, published in 2015, found that FIB four had an AUROC of only zero point six zero for detecting

cystic fibrosis liver disease, compared with zero point seven five for APRI. But FIB four had one strength. At the time cystic fibrosis liver disease was diagnosed, it detected portal hypertension with an AUROC of zero point nine one. This makes sense. When the platelets fall because of a big spleen, any score that has the platelets in the denominator will rise.

The AASLD guideline of 2024 says this clearly. FIB four performs worse in children, "particularly very young children". The general lesson is this. Any score that contains age, including FIB four, the adult NAFLD fibrosis score and others, needs its own paediatric cut-offs, or it should not be used in children at all.

Part three. What moves APRI in a child.

APRI has no age term, so it does better than FIB four in children. But it is made of two numbers, the AST and the platelet count, and each of them has its own biology outside the liver. Let us look at each side in turn.

First, the AST side. AST is not specific to the liver. It is also found in muscle and in red blood cells. So a child with a muscle disease, a child who has just had a seizure or hard exercise, or a blood sample that has haemolysed, will have a high AST and a high APRI without any liver injury. A rare cause is macro-AST, where AST is bound to an antibody and is cleared slowly from the blood. In each of these cases APRI will be high, but the liver is fine.

The AST also rises with liver inflammation. A child with a flare of autoimmune hepatitis, or with acute hepatitis, will have a very high AST, and the APRI will look like cirrhosis. This is the same trap we met with elastography in the last chapter. Inflammation looks like scar. So do not calculate a fibrosis score during a flare.

Second, the platelet side. Inflamed infants often have a reactive thrombocytosis, with platelet counts well above normal. This pulls the APRI down and can hide significant disease. On the other hand, when the spleen enlarges because of portal hypertension, the platelets fall, and the APRI rises. In that situation APRI is really reading portal hypertension, not fibrosis. And if the platelets are low for another reason, such as chemotherapy, sepsis, or immune thrombocytopenia, the APRI cannot be interpreted at all.

Third, there is the upper limit of normal for AST. Remember that APRI divides the AST by the laboratory's upper limit of normal. Different laboratories use different upper limits. If one laboratory uses an upper limit of thirty units per litre and another uses forty, the same child will have an APRI one third higher in the first laboratory. So before you compare an APRI from one hospital with a cut-off from a paper, find out which upper limit was used. The Cystic Fibrosis

Foundation guidance of 2024 recalculated its cut-offs using an upper limit of thirty for exactly this reason.

Finally, what does APRI actually track in the liver? A Finnish study from Helsinki by Nordenheim and colleagues, published in 2026, looked closely at this in biliary atresia. At the time of the Kasai operation, APRI barely correlated with the fibrosis stage, with a correlation coefficient of only zero point one six. What it did reflect was the ductular reaction, the proliferation of small bile ductules that is typical of biliary obstruction. After the Kasai operation, the picture changed, and APRI correlated reasonably with the METAVIR fibrosis stage, with a coefficient of zero point four six. And in both periods, APRI predicted the need for transplantation better than it staged the fibrosis.

Part four. Biliary atresia: APRI at Kasai, and MMP seven for the diagnosis.

Biliary atresia is the disease in which serum markers have been studied most in children. It is worth separating two very different questions. The first question is diagnostic. Does this cholestatic infant have biliary atresia? The second question is about severity and prognosis. In an infant who does have biliary atresia, how much fibrosis is there, and how will the child do?

Let us start with APRI at the time of the Kasai operation. In a British study by Grieve and colleagues, published in 2013, two hundred and sixty infants had APRI measured at the time of the operation. An APRI of one point two two or more predicted a liver that looked cirrhotic to the surgeon, with a sensitivity of seventy-five percent and a specificity of eighty-four percent. And infants in the lowest quarter of APRI values, below zero point four three, were the only group with better survival with their native liver.

A smaller Finnish study by Suominen and colleagues, published in 2015, looked at twenty-nine infants. The median APRI at the time of Kasai was one point three four in those who later needed a transplant, and zero point seven seven in those who kept their own liver.

After the Kasai operation, APRI does better at detecting cirrhosis. A meta-analysis by He and colleagues, published in 2021, pooled the post-Kasai studies. The pooled sensitivity was seventy-eight percent, the pooled specificity was eighty-three percent, and the area under the curve was zero point eight seven.

But here is the problem. Across fourteen studies in biliary atresia, the cut-off for F three fibrosis ranged from zero point seven to two point two six. That is a threefold range. No single cut-off transfers from one centre to another. And in infants with cholestasis that is not biliary atresia, a study by Brits and colleagues, published in 2025, found that an APRI cut-off of one point three eight caught

every infant with cirrhosis, but its specificity was only twenty-one percent. In other words, most infants above the cut-off did not have cirrhosis.

So APRI in biliary atresia is a prognostic signal more than a stage. And APRI should never be used to decide whether an infant has biliary atresia in the first place. For that question there is a much better blood test, MMP seven.

MMP seven is released by injured bile duct cells. In biliary atresia, the bile duct cells are under heavy attack, and serum MMP seven rises very high. A meta-analysis by Ma and colleagues, published in 2025, pooled seventeen studies. MMP seven had a sensitivity of ninety-three percent and a specificity of ninety percent for biliary atresia, compared with other causes of infant cholestasis. In a large study from Shanghai by Jiang and colleagues, published in 2019, two hundred and eighty-eight infants were tested, and the sensitivity was ninety-five percent and the specificity was ninety-three percent.

Compare that with the older tests. A raised GGT, pooled across studies, had a sensitivity of about eighty percent and a specificity of about seventy-nine percent. Interleukin thirty-three, another candidate marker, had a sensitivity of seventy-seven percent and a specificity of eighty-five percent. MMP seven is clearly better than both.

MMP seven also tracks fibrosis to some extent. Its correlation with the biopsy fibrosis stage is about zero point four seven. And in a Finnish study by Kerola and colleagues, published in 2016, MMP seven in the bile duct tissue correlated with the METAVIR stage after Kasai, with a coefficient of zero point six one.

There are two caveats. First, MMP seven works best in infants younger than sixty days. Second, it works better in Eastern cohorts than in Western ones. In the meta-analysis, the diagnostic odds ratio was about one hundred and fifty in Eastern studies and about fifty-one in Western studies. And the assay and the cut-off are not standardised. So each centre needs to validate its own cut-off before relying on it.

In practice, a high MMP seven raises the urgency of the work-up. It does not replace the cholangiogram, and a low MMP seven should never delay it.

Part five. GPR and cystic fibrosis.

Cystic fibrosis is the one disease where serum fibrosis scores have earned a real place in routine paediatric practice. There are good reasons for this. Cystic fibrosis liver disease is patchy, so a biopsy can easily miss it. Elastography is helpful but not available everywhere. And children with cystic fibrosis have blood tests at least once a year anyway.

Let us start with APRI. In the Houston study by Leung and colleagues, sixty-seven children with cystic fibrosis liver disease, confirmed by biopsy, were compared with one hundred and four children with cystic fibrosis and no liver disease. APRI had an AUROC of zero point seven five. An APRI above zero point two six four flagged liver disease with a sensitivity of seventy-three percent and a specificity of seventy percent.

In a second study, from Brisbane, by Lewindon and colleagues, published in 2019, the children had a dual-pass liver biopsy. This means two needle passes were taken, to reduce the chance of missing patchy disease. APRI had an AUROC of zero point eight one for F three to F four fibrosis. And when APRI was combined with liver stiffness on elastography, the AUROC for telling F three to F four from F one to F two rose to zero point nine two, with a specificity of one hundred percent.

Now GPR. Remember that GPR is the biliary twin of APRI, with GGT in place of AST. Since cystic fibrosis liver disease starts in the bile ducts, GPR makes biological sense. In an Australian study by Calvopina and colleagues, published in 2022, two hundred and thirty-seven children with cystic fibrosis were studied. GPR had an AUROC of zero point eight one for cystic fibrosis liver disease, at a cut-off of zero point two zero. In fifty-four children with biopsy-proven disease, GPR had an AUROC of zero point eight two for moderate to severe fibrosis, F two to F four, at a cut-off of zero point three two. And a GPR above zero point eight four at the time of diagnosis predicted portal hypertension, with an AUROC of zero point eight one.

GPR has also been tested in autoimmune hepatitis. In a Chinese study by Zhao and colleagues, published in 2026, fifty-two children with autoimmune hepatitis were studied, and GPR had an AUROC of zero point seven zero for advanced fibrosis. That is only fair.

The Cystic Fibrosis Foundation guidance of 2024 comments that GPR may detect early biliary disease before liver cells are injured. But remember one important limit. In the first months of life, GGT is physiologically several times higher than the adult upper limit of normal. So GPR is meaningless in young infants.

What does the guidance tell us to do? The Cystic Fibrosis Foundation recommendation, which received twenty-one out of twenty-two votes, is to calculate at least one fibrosis index, such as APRI, GPR or FIB four, and to follow it at least once a year. There is no accepted cut-off specific to cystic fibrosis. In practice, the trend matters more than any single value. And a falling platelet count signals portal hypertension, whatever the absolute value of the score.

Part six. Fatty liver and the ELF test.

Fatty liver disease, now called metabolic dysfunction-associated steatotic liver disease, or MASLD, is the most common liver disease in children. In adults with MASLD, FIB four is the standard first test. So it is natural to ask whether the same approach works in children.

The answer depends very much on which study you read. Small single-centre studies make some markers look almost perfect. Larger multicentre studies show what really survives.

Let us start with the impressive results. A study from Rome by Mosca and colleagues, published in 2019, found that P three NP alone had an AUROC of zero point nine nine three for advanced fibrosis, in two hundred and four children. An earlier study from Rome by Nobili and colleagues, published in 2009, found that the ELF test had an AUROC of zero point nine nine zero, in one hundred and twelve children. These numbers are close to perfect, and they come from a single expert centre.

Now the realistic results. In the ESPGHAN special interest group cohort, the study by Kalveram and colleagues that we met earlier, four hundred and fifty-seven children from ten European centres had a biopsy. For advanced fibrosis, APRI had an AUROC of zero point seven five nine. The Paediatric NAFLD Fibrosis Score had zero point seven one two, although in its original derivation study by Alkhouiri and colleagues it had zero point seven four zero. FIB four had zero point six one one. And the Paediatric NAFLD Fibrosis Index had zero point five two one, which is no better than tossing a coin.

The most striking finding of that study was that no score did better than a plain ALT above fifty units per litre. In other words, the complex scores added nothing to a simple liver enzyme. In a separate study by Mansoor and colleagues, APRI detected any fibrosis with an AUROC of zero point eight zero, but any fibrosis is not the clinically important question.

What about the ELF test? It looked excellent in Rome. But in children, growth looks like scar. Hyaluronic acid and P three NP rise during growth and peak at puberty, whether or not the child has liver disease. In a study by Thusing and colleagues, published in 2022, three hundred and forty-three healthy children aged six to seventeen years had an ELF test. Their median ELF was eight point nine. Only about six percent of these healthy children fell inside the adult reference range. Most healthy teenagers sit above the adult normal.

ELF can also miss real disease. In a small study of children with autosomal recessive polycystic kidney disease, in whom congenital hepatic fibrosis is

expected, the ELF test was normal even in children with possible or proven liver fibrosis.

Despite this, the NICE guideline NG forty-nine, updated in July 2026, uses an ELF of ten point five one or more to diagnose advanced fibrosis in children as well as adults, and recommends repeating the test every two years in children. This cut-off comes from the single Rome study. AASLD 2025 and ESPGHAN 2026 both say that ELF is not validated in children.

So in a teenager with fatty liver, the practical approach is simple. Measure the ALT, and if fibrosis needs to be assessed, use elastography or MR elastography. Do not reassure a teenager on the basis of a FIB four below one point three zero, because, as we saw, the age term pulls every teenager's FIB four down.

Part seven. Other diseases, and predicting the future.

Let us briefly go through the other diseases in which serum scores have been tested in children.

In children with a Fontan circulation, the liver is chronically congested. In an American study by Emamaullee and colleagues, published in 2021, one hundred and six adolescents with a Fontan circulation had a liver biopsy. Both APRI and FIB four had an AUROC of only zero point six nine for advanced fibrosis. In a smaller mixed cohort studied by Jarasvaraparn and colleagues in 2024, the nineteen children showed no discrimination at all. So the evidence is weak.

In chronic hepatitis B, two biopsy studies, one of seventy-one Polish children and one of one hundred and sixteen Chinese children, found that APRI above zero point five nine had an AUROC of about zero point seven five, and FIB four had zero point six zero for F two or more. Again, the evidence is weak.

In hepatitis C, a study of forty-eight Egyptian children found that APRI had an AUROC of zero point four nine for F two or more. That is no better than chance. And in the study of fifty-two Chinese children with autoimmune hepatitis, APRI and FIB four did not discriminate between fibrosis stages.

So outside cystic fibrosis and biliary atresia after Kasai, serum scores are weak staging tools in children.

But there is better news when we ask a different question. Serum scores are better at predicting the child's future than at staging the liver today. The reason is the platelet count. The platelets carry information about portal pressure, and clinical outcomes follow the portal pressure.

In biliary atresia, the Helsinki study by Nordenheim and colleagues found that each one-unit rise in APRI at the time of Kasai increased the risk of transplantation by forty percent, with a hazard ratio of one point four, in one hundred and sixteen infants. After Kasai, the median APRI was zero point seven in children who kept their native liver, and two point one in those who later needed a transplant.

In the Fontan circulation, a study by Martin de Miguel and colleagues, published in 2023, followed one hundred and fifty-nine adults after the Fontan operation. Each one-unit rise in APRI was linked to a higher risk of death from any cause, with a hazard ratio of one point three one. And in Fontan patients who had a biopsy, the platelet count began to fall years before bridging fibrosis appeared.

The AASLD guideline of 2024 adds a note of caution. There is not enough evidence to use serum scores to measure a change in fibrosis in children. So use the trend to decide when to do an endoscopy or when to refer, but do not use it to claim that fibrosis has regressed.

Part eight. What the guidelines say.

Let us bring the guidelines together. They do not all agree.

The AASLD guideline on blood-based assessment, published in 2025 and written by Sterling and colleagues, covers all paediatric chronic liver disease. It suggests that APRI or FIB four may be used to detect advanced fibrosis or cirrhosis in children. This suggestion is ungraded and based on low-quality evidence. The same guideline says that serum scores should not be used for monitoring change in children, and that the evidence is insufficient for fatty liver and for alpha one antitrypsin deficiency.

The AASLD paediatric fatty liver guidance of 2025 takes a stricter view. The adult scores perform poorly in children, and their cut-offs should not be applied. ELF is not validated.

The ESPGHAN position paper of 2026 on paediatric fatty liver agrees. FIB four, APRI, the Paediatric NAFLD Fibrosis Index and the Paediatric NAFLD Fibrosis Score are not recommended, with one hundred percent agreement among the authors. The evidence for ELF is insufficient. So for fatty liver in children, North America and Europe both say no to serum scores.

The NICE guideline NG forty-nine, updated in 2026, is the exception. It covers adults and children with fatty liver, and it uses an ELF of ten point five one or more to diagnose advanced fibrosis in children, with a repeat test every two years. This is a United Kingdom position, and it is not shared by AASLD or ESPGHAN.

The Cystic Fibrosis Foundation guidance of 2024 covers all ages. It suggests calculating at least one fibrosis index, such as APRI, GPR or FIB four, every year, while admitting that there is no accepted threshold for cystic fibrosis.

And the World Health Organization guideline on hepatitis B, published in 2024, uses APRI to decide who needs treatment when elastography is not available. An APRI above zero point five means significant fibrosis, and above one point zero means cirrhosis. The guideline applies these cut-offs to adults and adolescents. But the meta-analysis behind them, by Liguori and colleagues, excluded studies done only in children. So the adolescent recommendation is extrapolated from adults.

In summary, there are only two places where a guideline actively supports serum scores in children: cystic fibrosis, and the general, ungraded AASLD suggestion. For fatty liver, both major paediatric bodies say no, and only NICE says yes, to the ELF test.

Putting it into practice.

Let us finish with eight common situations.

A cholestatic infant who might have biliary atresia. Measure the GGT, and MMP seven if your laboratory has a validated assay. A high MMP seven raises the urgency. Do not use APRI or FIB four to make the diagnosis, and never delay the cholangiogram because of a low value.

An infant with biliary atresia at the time of the Kasai operation. Calculate APRI and read it together with what the surgeon sees. An APRI above about one point two suggests cirrhosis, and an APRI below about zero point four three carries the best outlook. Do not read it as a fibrosis stage.

A child with biliary atresia after the Kasai operation. Follow the APRI, the platelets and the spleen size every year. A rising APRI or falling platelets should prompt an endoscopy for varices. Do not call regression of fibrosis from a single value.

A child with cystic fibrosis. Calculate APRI or GPR every year, together with elastography where available. An APRI above about zero point two six, or a GPR above zero point two zero, means you should look harder for liver disease. Do not rely on FIB four to detect it.

A teenager with fatty liver. Measure the ALT, and use elastography or MR elastography to look for fibrosis. No serum score does better than an ALT above fifty. Do not reassure the teenager with a FIB four below one point three zero.

An adolescent with hepatitis B, where elastography is not available. Use APRI according to the WHO 2024 guideline. An APRI above zero point five flags eligibility for treatment. Do not extend this rule to young children.

An adolescent with a Fontan circulation. Follow the platelet count and the APRI over time, together with imaging. Falling platelets over the years are the warning sign. Do not exclude bridging fibrosis on the basis of one APRI.

And a child presenting with autoimmune hepatitis or acute hepatitis. Wait for remission. Then assess fibrosis, preferably with elastography. Do not calculate a fibrosis score during the flare.

Things to remember.

APRI, FIB four and GPR were all built in adults with viral hepatitis. Only the Paediatric NAFLD Fibrosis Score and the Paediatric NAFLD Fibrosis Index were derived in children.

FIB four multiplies by age. A fifteen-year-old with cirrhotic blood tests can read below the adult rule-out cut-off.

APRI divides by the laboratory's upper limit of normal for AST. Know the denominator before you compare numbers.

In biliary atresia, APRI at the time of Kasai reads ductular reaction and predicts transplantation. MMP seven, not APRI, helps make the diagnosis.

Cystic fibrosis is the one disease where a yearly APRI or GPR is guideline practice.

In paediatric fatty liver, no score beats an ALT above fifty units per litre. AASLD and ESPGHAN both say do not use the scores.

ELF rises with growth. Healthy teenagers sit above the adult normal, yet NICE still applies a cut-off of ten point five one in children.

A falling platelet count is the most honest serum marker. It tracks portal pressure.

Questions to test yourself.

Question one. A teenager and a middle-aged adult have exactly the same AST, ALT and platelet count. Why might FIB four suggest advanced fibrosis in the adult and reassure you in the teenager? Answer: because FIB four multiplies by age in years. With an AST of eighty, an ALT of sixty-four and platelets of one hundred and twenty, FIB four is three point seven five at forty-five years, above

the rule-in cut-off, but one point two five at fifteen years, below the rule-out cut-off.

Question two. Two laboratories report APRI for the same child, but one uses an upper limit of normal for AST of thirty and the other uses forty. How will the results differ? Answer: the laboratory using thirty will report an APRI about one third higher, because APRI divides the AST by the upper limit of normal.

Question three. Name three reasons, other than liver fibrosis, why APRI might be high in a child. Answer: a high AST from muscle injury, haemolysis or macro-AST; a high AST from active inflammation, such as an autoimmune hepatitis flare; and a low platelet count from chemotherapy, sepsis or immune thrombocytopenia.

Question four. What does APRI reflect at the time of the Kasai operation, according to the Helsinki study by Nordenheim and colleagues? Answer: the ductular reaction, not the fibrosis stage. The correlation with fibrosis was only zero point one six at Kasai, but rose to zero point four six after Kasai. APRI predicted transplantation better than it staged fibrosis.

Question five. What is the best serum test for diagnosing biliary atresia, and how accurate is it? Answer: MMP seven. In a meta-analysis of seventeen studies, it had a sensitivity of ninety-three percent and a specificity of ninety percent. It works best below sixty days of age, and each centre should validate its own cut-off.

Question six. What does the Cystic Fibrosis Foundation 2024 guidance recommend about serum fibrosis scores? Answer: calculate at least one fibrosis index, such as APRI, GPR or FIB four, and follow it at least once a year. There is no accepted cut-off specific to cystic fibrosis.

Question seven. In the ten-centre European study of four hundred and fifty-seven children with fatty liver, which test did the serum scores fail to beat? Answer: a plain ALT above fifty units per litre. APRI had an AUROC of zero point seven six and FIB four only zero point six one for advanced fibrosis.

Question eight. Why is the ELF test hard to interpret in children, and which guideline still uses it? Answer: hyaluronic acid and P three NP rise with growth and peak at puberty, so healthy children have a median ELF of about eight point nine and most sit above the adult reference range. NICE NG forty-nine still uses a cut-off of ten point five one in children, while AASLD 2025 and ESPGHAN 2026 say ELF is not validated.

End of chapter.